

Measuring Hierarchy with Multiple Supervisors

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Abstract

This paper develops an axiomatic framework for measuring hierarchical structures in modern organizations, where multiple supervisors and non-traditional arrangements, such as matrix structures, are common. Representing hierarchies as directed acyclic graphs (DAGs), we show that structural flattening in these organizations is inherently path-dependent: due to chronological interference, determining hierarchical dominance generally requires explicitly tracing the sequence of subordination removals. We show that this complexity resolves in multipath-free hierarchies. Without redundant reporting lines, the chronological order of removals leaves a permanent static footprint, allowing hierarchical dominance to be verified by a single geometric condition rather than a step-by-step procedure. Our axiomatic framework is grounded in Anonymity, the Replication Principle, and a bifurcated Subordination Removal postulate. The distinction between baseline and Strong Subordination Removal clarifies whether redundant formal reporting lines strictly contribute to hierarchical weight or instead leave the effective chain of command unchanged. Using this foundation, we characterize the class of hierarchical pre-orders consistent with sequential subordination removal, providing a systematic basis for both incomplete hierarchy comparisons and complete extensions across organizations of varying sizes.

Keywords: hierarchy measurement, hierarchical pre-order, hierarchical index, directed acyclic graph.

JEL classifications: D23, L22.

Practitioner point: Axiomatic measurement of complex organizational hierarchies.

1. Introduction

Organizational hierarchical structures are not merely circumstantial; they are closely linked to income distribution, economic power concentration, and systemic inequality. Therefore, understanding and systematically measuring hierarchical structures is crucial for analyzing the power dynamics shaping resource distribution and individual and collective economic outcomes.

The correlation between hierarchical rank and income is remarkably consistent across diverse organizations. As evidenced by case studies, income scales strongly with hierarchical rank within firms, exerting a more significant influence on earnings than any other

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measurable factor. This relationship follows a predictable pattern, where income increases exponentially with rank, forming the foundation for power-law distributions observed among top incomes across the economy (Fix, 2018, 2021).

Research also suggests that societies with higher energy consumption tend to develop larger, more hierarchical institutions (Fix, 2017). Hierarchical complexity rises with energy, concentrating resources at the top. By examining the “energy-hierarchy-inequality” relationship, Fix (2019) traces the origins of inequality, revealing how hierarchical structures emerged alongside increasing energy consumption and technological advancement.

From a theoretical perspective, Simon (1957) and Lydall (1959) pioneered models demonstrating how hierarchical structures generate power-law income distributions, which characterize economic inequality. In a stylized framework with a fixed ratio between the number of supervisors in each grade and the number of immediate subordinates, they showed that when two exponential trends—income growth with hierarchical rank and decreasing positions at higher ranks—intersect, a power-law distribution naturally emerges.

As an initial step toward systematically exploring the theoretical connections between hierarchy and the determinants of power and resource distribution, Carbonell-Nicolau (2025) established a foundational framework for measuring hierarchy, specifically tailored to tree-like hierarchical structures. The present paper builds on this methodology, extending the analysis to encompass more complex hierarchical systems. Our work broadens the scope to include hierarchies where individuals may have multiple immediate supervisors, a characteristic prevalent in modern hierarchies such as matrix organizational structures.

Matrix organizations, widely adopted across various industries, including IT, consulting, aerospace, and defense, are characterized by employees reporting to both functional and project managers. According to Gallup (2017), 84% of U.S. employees work in organizations utilizing matrixed arrangements to varying degrees.

We use *directed acyclic graphs* (DAGs) to represent these complex hierarchical frameworks. These graphs, where nodes represent individuals within the hierarchy and directed edges denote supervisor-subordinate relationships, are versatile objects encompassing various organizational structures. They can accommodate multiple direct reports and supervisors while ensuring acyclicity to prevent circular reporting chains.

Directed acyclic graphs have gained prominence across disciplines for their ability to model ordered relationships with clear, unidirectional reporting structures. In the economic analysis of the firm, they serve as the natural mathematical representation of the formal organizational chart, capturing the explicit distribution of authority and decision-making power.

While actual workplace dynamics may also feature cyclical feedback mechanisms or informal communication channels—what Simon (1981) termed “informal organizations,” whose hierarchical tendencies have been quantified by Krackhardt (1994) (see also Everett and Krackhardt, 2012)—these fall outside the scope of our analysis. Our measurement approach focuses squarely on the explicit, formal subordination structures that dictate institutional resource allocation.

By anchoring our analysis in these versatile acyclic structures, we are able to develop a rigorous axiomatic framework for comparing organizational complexity. We evaluate these hierarchies through the lens of fundamental postulates: Anonymity, Subordination Removal, Strong Subordination Removal, and the Replication Principle. Anonymity and the Replication Principle were introduced in Carbonell-Nicolau (2025). Anonymity ensures

label-independence, meaning that renaming individuals in a hierarchy leaves its structural properties unchanged. The Replication Principle maintains consistency across scales by asserting that creating “organizational clones”—exact replicas of a hierarchy—preserves its hierarchical character.

The core of our approach rests on the Subordination Removal postulate, which formalizes the intuitive principle that eliminating a reporting line flattens the organization. However, extending this concept to general DAGs introduces substantial topological complexity. In our framework, removing a formal subordination relation between a supervisor j and a subordinate i eliminates the direct edge while preserving the broader chain of command: if j has immediate superiors, those immediate superiors are connected directly to i , bypassing j , unless they already possess an alternative path to i that does not traverse j .

This bypass mechanism reveals a fundamental challenge unique to directed acyclic graphs: chronological interference. In general DAGs, overlapping matrix paths and redundant authority channels make the outcome of subordination removals depend on the order in which they are performed. The generation of bypass edges is evaluated in the current intermediate hierarchy, not merely in the initial graph. Consequently, determining whether one hierarchy can be reduced to another generally requires explicitly tracing the chronological sequence of subordination removals.

A central theoretical contribution of this paper is mapping the precise boundary where this chronological complexity gives way to geometric tractability. We demonstrate that by restricting the domain to *multipath-free hierarchies*—where matrix reporting is permitted but multiple directed paths between the same ordered pair of individuals are ruled out—chronological dependencies become statically visible. On this restricted domain, the existence of a valid sequence of subordination removals is equivalent to a clean, one-shot geometric condition based on reachability containment and unique-path consumption.

Furthermore, the transition to matrix structures introduces a natural bifurcation in the subordination-removal concept itself. By distinguishing between baseline Subordination Removal and Strong Subordination Removal, our dual-consistency framework clarifies how different measures treat formal reporting redundancy. Baseline Subordination Removal evaluates hierarchy through the effective, transitive chain of command, allowing the removal of a redundant formal edge to leave hierarchical weight unchanged when the supervisory reachability relation is unaffected. Conversely, Strong Subordination Removal treats every formal reporting line as a strict contributor to hierarchical weight, demanding that the elimination of any edge strictly flattens the hierarchy.

Using this axiomatic foundation, we characterize the class of hierarchical pre-orders that are consistent with sequential subordination removal. The analysis yields a hierarchy of measures ordered by their degree of completeness. These include an incomplete comparison based on the ordered preservation of supervisory relationships and a complete cardinal index based on the average number of supervisors per individual. Finally, the Replication Principle allows us to generalize our main characterizations from fixed-size organizations to hierarchies of varying sizes.

A substantial body of literature on hierarchy measurement spans multiple disciplines, offering diverse hierarchical indices that comprehensively rank pairs of hierarchies. These measures range from intuitive indicators—such as the count of organizational layers from top executive to lowest employee or the “span of control” quantifying the average number of direct reports per manager—to more sophisticated statistical approaches that

assess node asymmetries, directional relationships, and top-down reachability patterns in hierarchies and broader network structures (see, e.g., [Trusina et al., 2004](#); [Luo and Magee, 2011](#)). A notable example among these advanced measures is the “global reaching centrality” concept ([Mones et al., 2012](#)), which evaluates individuals’ positional centrality within hierarchies based on their capacity to reach other nodes. As demonstrated in [Carbonell-Nicolau \(2025\)](#), this centrality-based approach diverges fundamentally from the axiomatic framework we develop in this paper.

[Corominas-Murtra et al. \(2013\)](#) map directed graphs, not necessarily acyclic, onto a concise set of quantitative dimensions—Treeness, Feedforwardness, and Orderability—each capturing distinct aspects of hierarchical structure. However, unlike our framework, which differentiates among acyclic structures, their methodology assigns identical hierarchical degrees to all acyclic graphs, treating them as equivalent. In contrast, [Czégel and Palla \(2015\)](#) developed an approach with discriminatory power among acyclic graphs by employing a simulation technique. Their method reverses traditional information flow patterns by simulating random walks from employees upward to managers, thereby tracing information to its origins. Within this framework, nodes accumulating numerous walkers occupy superior positions in the hierarchy, while those attracting fewer walkers reside in subordinate positions. Their hierarchy score quantifies the distributional inequality of these walkers—greater concentration at upper echelons signifies a more pronounced hierarchical structure.

Our approach diverges from these algorithmic methods by anchoring hierarchy measurement in precisely articulated theoretical principles. We identify a core incomplete pre-order that captures the unambiguous comparisons implied by the fundamental axioms. This core pre-order serves as the foundation for partial and complete extensions that preserve these essential properties. Importantly, while enhancing a hierarchical measure’s completeness allows for ranking more hierarchies, this expanded capacity comes with a consequential trade-off: it involves making additional judgments in cases not fully resolved by the basic axioms. This shift introduces a tension between comprehensive applicability and theoretical discipline.

The paper is structured as follows. [Section 2](#) introduces the foundational concept of directed acyclic graphs, providing the necessary background for understanding hierarchical structures. [Section 3](#) establishes the mechanism of sequential subordination removals for fixed-size organizations, analyzes the geometric properties of the multipath-free domain, and presents the axiomatic characterizations of consistent pre-orders. The extension of this framework to hierarchies of varying sizes is detailed in [Section 4](#). Finally, [Section 5](#) concludes the paper, summarizing the main contributions and outlining directions for future research.

2. Hierarchies

We define a *hierarchy* as a finite *directed acyclic graph* (DAG). Thus, a hierarchy h consists of a finite set of nodes $V(h)$ and a set of directed edges $E(h) \subseteq V(h) \times V(h)$, with the property that no directed cycle exists. Acyclicity ensures that following directed edges can never lead from a node back to itself.

In our interpretation, nodes represent individuals, while directed edges represent immediate subordination relations. An edge $j \rightarrow i$ means that individual j is an immediate

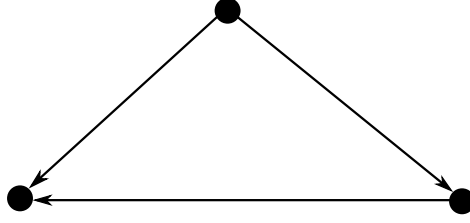


Figure 1: A hierarchy.

supervisor of individual i , or equivalently that i is an immediate subordinate of j . More generally, j is a supervisor of i if there exists a directed path of positive length from j to i . If this path has length one, then j is an immediate supervisor of i ; otherwise, j is an indirect supervisor of i .

For each individual $i \in V(h)$, let S_i denote the set of all supervisors of i , both immediate and indirect. Equivalently, S_i is the set of strict ancestors of i in h . Acyclicity implies a useful hierarchical constraint: if individual i has k supervisors, then every supervisor of i has strictly fewer than k supervisors.

Indeed, suppose that $j \in S_i$. Since the supervisory relation is transitive, every supervisor of j is also a supervisor of i , so $S_j \subseteq S_i$. Moreover, $j \in S_i$ but $j \notin S_j$, because acyclicity rules out self-supervision. Hence $S_j \subsetneq S_i$, and therefore

$$|S_j| < |S_i| = k.$$

Figure 1 illustrates a simple hierarchy comprising three individuals connected by three subordination relations.

This directed-acyclic-graph formulation is flexible enough to represent complex reporting structures in which an individual may have multiple immediate supervisors. It therefore accommodates matrix organizational structures and other non-traditional hierarchical arrangements while preserving the acyclicity of formal authority relations.

This approach differs from the hierarchy definition presented in [Carbonell-Nicolau \(2025\)](#), which constrained each individual to have at most one immediate supervisor and explicitly partitioned nodes into distinct ranks or levels. The present conceptualization is more general: it relies only on subordination relationships between individuals and does not impose predefined rank structures. Nevertheless, rank-based analysis remains compatible with the framework and can be incorporated when formal hierarchical levels are available.

For each individual i in a hierarchy h , we define the *sub-hierarchy* $h(i)$ as the subgraph of h induced by i and all of its direct and indirect subordinates. That is, $h(i)$ contains i , every node reachable from i by a directed path, and all directed edges among these nodes that are present in h .

This sub-hierarchy preserves the supervisory relationships below i and constitutes a hierarchy in its own right.

Notationally, a hierarchy comprising $m \in \{1, 2, \dots\}$ independent hierarchies $h_1, \dots, h_m$ is denoted by $(h_1, \dots, h_m)$. Here, *independence* means that the hierarchies are pairwise

disconnected: for any $r \neq s$, no directed edge exists between any node in h_r and any node in h_s .

Finally, we define the *size* of a hierarchy h as the cardinality of its node set, $|V(h)|$; that is, the total number of individuals represented within the hierarchical structure.

3. Hierarchical dominance for fixed-size organizations

We initially restrict our analysis to hierarchies of a fixed size n , deferring the examination of variable-sized hierarchies to a subsequent section.

For a hierarchy h , let $V(h)$ and $E(h)$ denote its node set and edge set, respectively. Let $\mathcal{H}_n$ denote the set of all hierarchies h with $|V(h)| = n$, representing the complete collection of n -person organizational structures that satisfy our directed acyclic graph definition.

A hierarchical pre-order $\succsim$ is defined as a reflexive and transitive binary relation on the set $\mathcal{H}_n$. These pre-orders represent potentially incomplete comparative assessments between pairs of hierarchies, where the notation $h \succsim h'$ carries the interpretation that “hierarchy h exhibits at least as much hierarchical structure as hierarchy h' .”

Definition 1. A hierarchical pre-order on $\mathcal{H}_n$ is a reflexive and transitive binary relation on $\mathcal{H}_n$.

The symmetric and asymmetric parts of a hierarchical pre-order $\succsim$ are denoted by $\sim$ and $>$, respectively.

The following property of a hierarchy, called Anonymity, was introduced in Carbonell-Nicolau (2025).

A hierarchy h' is said to be a *relabeling* of another hierarchy h if h' is obtained from h by relabeling the individuals in h . Formally, there is a bijection $\phi : V(h) \rightarrow V(h')$ preserving immediate subordination relations: for every pair of nodes $i, j \in V(h)$, j is an immediate supervisor of i in h if and only if $\phi(j)$ is an immediate supervisor of $\phi(i)$ in h' .

Anonymity (A). A hierarchical pre-order $\succsim$ on $\mathcal{H}_n$ satisfies **A** if for any two hierarchies h and h' in $\mathcal{H}_n$, $h \sim h'$ whenever h' is a relabeling of h .

This axiom emphasizes that the essential structure of a hierarchy remains invariant under relabeling.

The concept of subordination relation removal, originally introduced by Carbonell-Nicolau (2025), is extended in this work to align with our more flexible definition of a hierarchy.

Definition 2. We say that hierarchy h' is obtained from hierarchy h by *removing a subordination relation* if there exist a subordinate $i \in V(h)$ and an immediate supervisor j of i in h such that h' is obtained as follows:

- First delete the edge $j \rightarrow i$.
- If j has no immediate supervisors in h , no bypass edge is added.
- If j has at least one immediate supervisor in h , then for each immediate supervisor p of j in h , add the edge $p \rightarrow i$ unless there already exists in h a directed path from p to i that does not traverse j , i.e., a path on which j is not a node.

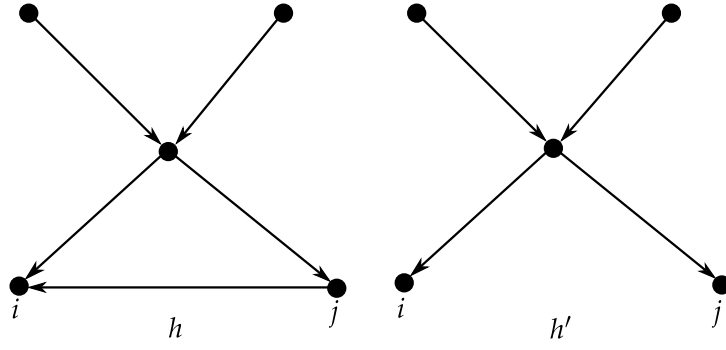


Figure 2: Subordination removal without a bypass edge.

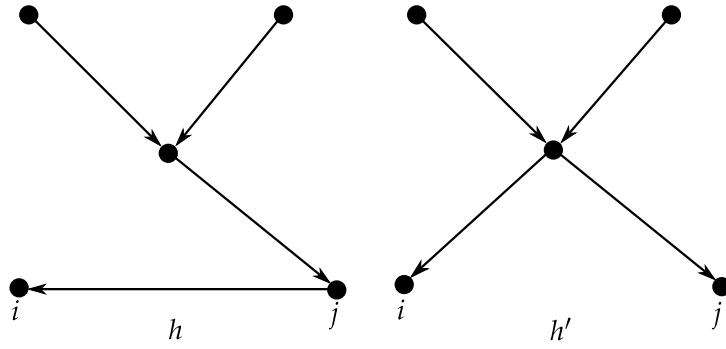


Figure 3: Subordination removal with a bypass edge.

- All other edges are preserved.

This definition formalizes subordination removal in two cases. When the supervisor j has no immediate supervisors, the link $j \rightarrow i$ is simply deleted. When j has immediate supervisors, each immediate supervisor p of j is connected directly to i , bypassing j , unless p already has an alternative route to i that does not pass through j . Thus, a bypass is added precisely when p 's authority over i would otherwise be mediated only through j . Each bypass $p \rightarrow i$ short-circuits a pre-existing path $p \rightarrow j \rightarrow i$, so the operation preserves acyclicity. Hence the resulting graph is again a hierarchy.

To illustrate this definition, consider the hierarchies from [Figure 2](#), where individual i has multiple direct supervisors, including j . We focus on the direct subordination relation $j \rightarrow i$. In this instance, h' is obtained from h by removing $j \rightarrow i$. No bypass edge is added for any immediate supervisor of j that already has a path to i not traversing j .

[Figure 3](#) presents a case where h' is also obtained from h by removing $j \rightarrow i$, but now a bypass edge must be added. After deleting $j \rightarrow i$, an immediate supervisor p of j would otherwise have no path to i except through j . Hence the edge $p \rightarrow i$ is added.

In [Figure 4](#), we remove the subordination relation $j' \rightarrow i$ by deleting the directed edge from j' to i . In this case, no bypass edge is added from the supervisors of j' to i , since those supervisors already have alternative paths to i that do not traverse j' .

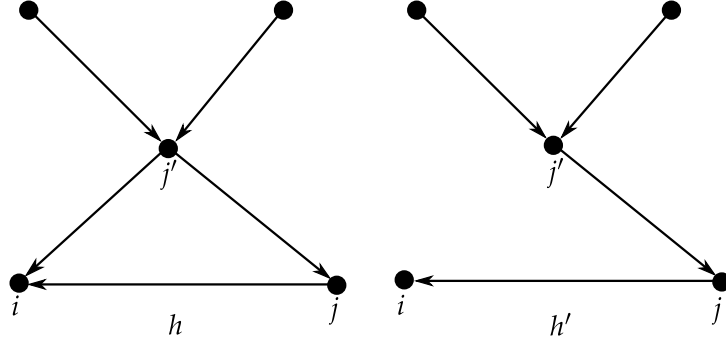


Figure 4: Subordination removal when alternative paths already exist.

The previous figures illustrate the three relevant possibilities under [Definition 2](#): simple deletion when no bypass is required, deletion with a new bypass edge, and deletion without bypasses because independent alternative paths already exist.

An edge $j \rightarrow i$ is *redundant* if there exists a directed path from j to i distinct from $j \rightarrow i$ itself.

We maintain the principle, originally introduced by [Carbonell-Nicolau \(2025\)](#), that the removal of a subordination relation results in a less hierarchical structure.

Subordination Removal (SR). A hierarchical pre-order $\succsim$ on $\mathcal{H}_n$ satisfies **SR** if for any two hierarchies h and h' in $\mathcal{H}_n$, $h \succsim h'$ whenever h' is obtained from h by removing a subordination relation. Moreover, $h \succ h'$ whenever the removed edge is non-redundant.

Thus, baseline Subordination Removal requires weak dominance for every removal, and strict dominance whenever the removed edge changes the transitive supervisory relation. The weak part is needed because deleting a redundant formal edge may leave the underlying reachability relation unchanged. Under [Definition 2](#), such a deletion may nevertheless generate bypass edges from immediate supervisors of j to i ; these bypasses preserve upstream authority without creating new reachability relations. Hence **SR** treats hierarchy primarily through the effective chain of command.

In some contexts, it is natural to view *any* formal reporting line—even a redundant one—as strictly contributing to the hierarchical weight of an organization. For instance, in matrix structures, redundant edges may represent dual reporting obligations, additional oversight, or bureaucratic rigidity. From this perspective, eliminating a reporting line should always strictly flatten the organizational structure, regardless of whether an alternative indirect chain of command remains intact. This motivates a stronger version of the Subordination Removal postulate.

Strong Subordination Removal (SSR). A hierarchical pre-order $\succsim$ on $\mathcal{H}_n$ satisfies **SSR** if for any two hierarchies h and h' in $\mathcal{H}_n$, $h \succ h'$ whenever h' is obtained from h by removing a subordination relation.

3.1. Sequential subordination removal and reduction traces

A central objective of our axiomatic framework is to identify a structural property of hierarchies that captures the operational process of successive subordination removals.

In the tree-based domain of Carbonell-Nicolau (2025), this was possible through a clean, one-shot geometric condition based on supervisory ranks and path containment.

For general directed acyclic graphs, however, a static topological characterization is obstructed by chronological interference. In a general DAG, matrix reporting allows for multiple, intersecting paths between nodes. Consequently, removing one reporting line may alter the path structure relevant for determining whether a later removal generates a bypass edge. Because bypass generation is evaluated in the *current* intermediate graph, the transformation is path-dependent: the order of removals may determine the final topology.

Accordingly, the substantive notion of reduction in the DAG setting is simply *sequential subordination removal*. For formal purposes, however, it is useful to record such a sequence explicitly. We do so by means of a *reduction trace*, which is nothing more than a step-by-step record of the removed edges together with the corresponding intermediate edge sets. The trace is not a new conceptual primitive; it is a bookkeeping device that makes chronological dependence explicit and will be useful below when identifying the boundary between general DAG complexity and the multipath-free case.

The formalization begins by aligning the node sets of the two hierarchies being compared. Since individuals in h and h' may be labeled differently, we use a bijection ϕ to translate the target graph h' back onto the nodes of the original graph h . We denote this “pulled-back” target edge set by E_ϕ^* .

To verify that h' is obtainable from h by sequential subordination removals, one records the removed edges together with the intermediate topological “snapshots” of the organization.

Definition 3 (reduction trace). Let $h, h' \in \mathcal{H}_n$, and let $\phi : V(h) \rightarrow V(h')$ be a bijection. Define the target edge set pulled back to h by

$$E_\phi^* = \{(u, v) : (\phi(u), \phi(v)) \in E(h')\}.$$

A *reduction trace* from h to h' via ϕ consists of a finite list of edges $e_1, \dots, e_L$ and a finite list of edge sets $E^0, E^1, \dots, E^L$ such that:

- (i) $E^0 = E(h)$;
- (ii) for each $\ell \in \{1, \dots, L\}$, the edge $e_\ell = j_\ell \rightarrow i_\ell$ belongs to $E^{\ell-1}$;
- (iii) E^ℓ is obtained from $E^{\ell-1}$ by deleting e_ℓ and adding exactly the bypass edges prescribed by Definition 2 applied to the intermediate graph $(V(h), E^{\ell-1})$;
- (iv) $E^L = E_\phi^*$.

Thus, a reduction trace is simply a precise record of a finite sequence of subordination removals after relabeling. Condition (i) fixes the starting hierarchy. Condition (ii) guarantees that an edge can only be removed if it is present in the current intermediate graph. Condition (iii) enforces that every step obeys Definition 2 relative to the *current* intermediate graph, thereby correctly tracking chronology-dependent bypass generation. Finally, condition (iv) requires that the resulting edge set coincide exactly with the pulled-back target hierarchy.

We now define the hierarchical dominance relation for general DAGs.

Definition 4 (DAG hierarchical dominance). For any two hierarchies $h, h' \in \mathcal{H}_n$, we write $h \succsim_H h'$ if and only if there exists a bijection $\phi : V(h) \rightarrow V(h')$ and a reduction trace from h to h' via ϕ .

Equivalently, $h \succsim_H h'$ holds if and only if, after pulling back labels through some bijection ϕ , the target hierarchy can be obtained from h by a finite sequence of DAG subordination removals.

As is standard for pre-orders, let $>_H$ and $\sim_H$ denote the asymmetric and symmetric parts of $\succsim_H$, respectively.

Two hierarchies h and h' are considered isomorphic, denoted $h \cong h'$, if there exists a bijection $\phi : V(h) \rightarrow V(h')$ such that for all individuals $i, j \in V(h)$,

$$j \rightarrow i \in E(h) \quad \Leftrightarrow \quad \phi(j) \rightarrow \phi(i) \in E(h').$$

We begin by verifying that the trace-based relation $\succsim_H$ is itself a well-defined hierarchical pre-order and that it respects relabeling.

Lemma 1. *The binary relation $\succsim_H$ defined on $\mathcal{H}_n$ is reflexive and transitive, and satisfies A.*

The proof of this lemma is given in [Appendix A.1](#). Having established the basic order-theoretic properties of $\succsim_H$, we next identify the effect of a single subordination removal on the induced strict relation.

Lemma 2. *For $h, h' \in \mathcal{H}_n$, if h' is obtained from h by a single subordination removal in the sense of [Definition 2](#), then $h >_H h'$.*

[Lemma 2](#) gives the one-step strictness implication. Its proof is relegated to [Appendix A.3](#).

To extend this from a single removal to arbitrary finite sequences, we need a simple monotone quantity that strictly decreases along every nonempty reduction trace. The following multiset provides such a tool and will allow us to rule out reverse weak dominance once a nonempty sequence of removals has occurred.

For any hierarchy g , let $s_g(x)$ denote the depth of node x , i.e., the cardinality of the set of strict ancestors of x in g . Define the multiset of source-node depths by

$$M_{\text{src}}(g) = \{ s_g(x) : (x, y) \in E(g) \}.$$

Lemma 3. *If there is a nonempty reduction trace from h to h' , then*

$$M_{\text{src}}(h') < M_{\text{src}}(h).$$

The proof is deferred to [Appendix A.2](#).

With this descent property in hand, we can now pass from the one-step case to a full characterization of strict dominance.

Proposition 1. *For hierarchies $h, h' \in \mathcal{H}_n$, $h >_H h'$ if and only if h' can be obtained from some relabeling of h by successive DAG subordination removals of one or more subordination relations.*

Proof. Sufficiency. Suppose that h' is obtained from a relabeling $\widehat{h}$ of h by a sequence of one or more subordination removals. Let $\rho : V(h) \rightarrow V(\widehat{h})$ be the relabeling bijection.

Transporting each removed edge and each intermediate edge set in this sequence back along ρ^{-1} , and using the fact that [Definition 2](#) commutes with relabeling, yields a reduction trace from h to h' via the corresponding target-labeling bijection. Hence $h \succ_H h'$.

Because the sequence contains one or more removals, the corresponding trace is nonempty. By [Lemma 3](#),

$$M_{\text{src}}(h') < M_{\text{src}}(h).$$

We claim that $h' \not\succ_H h$. Suppose, toward a contradiction, that $h' \succ_H h$.

If the trace from h' to h is nonempty, then [Lemma 3](#) gives

$$M_{\text{src}}(h) < M_{\text{src}}(h'),$$

contradicting the asymmetry of the strict multiset order.

If the trace from h' to h is empty, then h' is a relabeling of h . Relabeling preserves ancestor relations and hence source depths, so

$$M_{\text{src}}(h') = M_{\text{src}}(h),$$

again contradicting

$$M_{\text{src}}(h') < M_{\text{src}}(h).$$

Therefore $h' \not\succ_H h$. Since $h \succ_H h'$ and $h' \not\succ_H h$, we conclude that $h >_H h'$.

Necessity. Suppose $h >_H h'$. Then $h \succ_H h'$. By [Definition 4](#), there exists a reduction trace from h to h' via some bijection $\phi : V(h) \rightarrow V(h')$. The explicit list $e_1, \dots, e_L$ and transitions $E^{\ell-1} \rightarrow E^\ell$ realize a valid finite sequence of subordination removals from h to the hierarchy $\phi^{-1}(h')$ defined on the node set $V(h)$. Because [Definition 2](#) commutes with relabeling, applying ϕ to every intermediate hierarchy in this sequence yields a valid sequence of subordination removals from the relabeling $\phi(h)$ of h exactly to h' .

It remains to show that $L > 0$. If $L = 0$, then

$$E(h) = \phi^{-1}(E(h')),$$

so h' is a relabeling of h . Hence the inverse relabeling gives an empty trace from h' to h , implying $h' \succ_H h$. This contradicts $h >_H h'$, the asymmetric part of $\succ_H$. Therefore $L > 0$.

Hence h' is obtained from a relabeling of h by successive removals of one or more subordination relations. ■

3.1.1. The limits of static topological characterizations

Because the reduction trace in [Definition 3](#) relies on a chronological sequence of intermediate subgraphs, it diverges from the purely static, one-shot geometric characterization of hierarchical dominance established for trees in [Carbonell-Nicolau \(2025\)](#). To understand why this sequential formalization is mathematically necessary for general DAGs, it is useful to first examine a single, isolated subordination removal.

When restricted to exactly one operation, the subordination removal from [Definition 2](#) can be perfectly captured by a static topological condition. This one-step formulation mirrors the geometric aesthetic of the H -dominance condition in [Carbonell-Nicolau \(2025\)](#), adapted to accommodate multiple intersecting reporting lines.

Definition 5 (one-step topological reduction). Let $h, h' \in \mathcal{H}_n$. We say that h' is a *one-step topological reduction* of h if there exists a bijection $\phi : V(h) \rightarrow V(h')$ and an immediate subordination relation $j \rightarrow i \in E(h)$ satisfying the following conditions:

- (i) For each individual $x \in V(h) \setminus \{i\}$, an individual y' is an immediate supervisor of $\phi(x)$ in h' if and only if $\phi^{-1}(y')$ is an immediate supervisor of x in h .
- (ii) For the individual i , an individual y' is an immediate supervisor of $\phi(i)$ in h' if and only if $\phi^{-1}(y')$ is either:
 - (a) an immediate supervisor of i in h distinct from j , or
 - (b) an immediate supervisor of j in h such that every directed path from $\phi^{-1}(y')$ to i in h traverses j .

This definition translates the algorithmic procedure of edge deletion and bypass generation into a static evaluation of pre-images under ϕ^{-1} . Condition (i) ensures that the supervisory structure for all individuals other than the target subordinate i remains invariant. Condition (ii) governs the immediate supervisors of i , ensuring that all pre-existing supervisors other than j are retained, while specifying the exact topological conditions under which an immediate supervisor of j is granted bypass authority over i .

Proposition 2. For hierarchies $h, h' \in \mathcal{H}_n$, h' is a one-step topological reduction of h if and only if h' is a relabeling of a hierarchy obtained from h by a single subordination removal in the sense of [Definition 2](#).

Proof. Necessity. Suppose h' is a one-step topological reduction of h via the bijection ϕ , witnessed by the edge $j \rightarrow i \in E(h)$. Let $\hat{h}$ be the hierarchy obtained from h by applying [Definition 2](#) to remove $j \rightarrow i$. We show that ϕ is an isomorphism from $\hat{h}$ to h' .

By [Definition 2](#), the only possible changes from h to $\hat{h}$ are among edges entering i : the edge $j \rightarrow i$ is deleted, all other edges whose target is not i are preserved, and bypass edges $p \rightarrow i$ are added exactly for immediate supervisors p of j such that there is no directed path from p to i in h avoiding j .

Condition (i) in [Definition 5](#) states precisely that, for every node $x \neq i$, the immediate supervisors of $\phi(x)$ in h' are exactly the ϕ -images of the immediate supervisors of x in h . Hence all edges not entering i are preserved under ϕ .

For the target node i , Condition (ii)(a) says that every immediate supervisor of i in h other than j remains an immediate supervisor of $\phi(i)$ in h' . The deleted edge $j \rightarrow i$ is absent from h' , since j is neither an immediate supervisor of i distinct from j , nor an immediate supervisor of j , by acyclicity.

Finally, Condition (ii)(b) says that an edge $\phi(p) \rightarrow \phi(i)$ is present in h' exactly when p is an immediate supervisor of j in h and every directed path from p to i in h traverses j . This is equivalent to saying that no directed path from p to i avoids j , which is exactly the bypass condition in [Definition 2](#). Therefore the incoming edges of $\phi(i)$ in h' are exactly the ϕ -images of the incoming edges of i in $\hat{h}$.

Thus ϕ preserves and reflects every edge of $\hat{h}$, so h' is a relabeling of $\hat{h}$. Hence h' is a relabeling of a hierarchy obtained from h by a single subordination removal.

Sufficiency. Suppose h' is a relabeling of a hierarchy $\hat{h}$ obtained from h by a single subordination removal of the edge $j \rightarrow i$. Since $\hat{h}$ is obtained from h by modifying only

the edge set, we identify $V(\widehat{h})$ with $V(h)$. Let $\phi : V(h) \rightarrow V(h')$ be the relabeling bijection from $\widehat{h}$ to h' .

By **Definition 2**, no edges are modified except edges entering i : the edge $j \rightarrow i$ is deleted, all pre-existing edges $x \rightarrow i$ with $x \neq j$ are preserved, and bypass edges $p \rightarrow i$ are added exactly for immediate supervisors p of j such that every directed path from p to i in h traverses j .

Therefore, for every $x \neq i$, the immediate supervisors of $\phi(x)$ in h' are exactly the ϕ -images of the immediate supervisors of x in h , which gives Condition (i). For i , the immediate supervisors of $\phi(i)$ in h' are exactly the ϕ -images of nodes satisfying either Condition (ii)(a) or Condition (ii)(b). Hence h' is a one-step topological reduction of h . ■

Non-commutativity and path redundancy

Mathematically, the process of sequential subordination removal is the finite composition of the one-step topological reduction characterized in **Definition 5**. To encapsulate such a composition within a single static topological condition—analogueous to the H -dominance characterization of **Carbonell-Nicolau (2025)**—one must be able to reverse-engineer the transformation from h to h' from the initial and final graphs alone.

In directed acyclic graphs, we face a dual challenge: removals may fail to commute, and path redundancy dynamically alters how bypasses are generated. These are separate but compounding problems.

To see that non-commutativity occurs even in the absence of path redundancy, consider a simple four-person chain of command, h , with directed edges $a \rightarrow b$, $b \rightarrow c$, and $c \rightarrow d$. Suppose a restructuring initiative requires the removal of two specific reporting lines: $b \rightarrow c$ and $c \rightarrow d$.

If we remove $b \rightarrow c$ first:

1. The deletion of $b \rightarrow c$ triggers a bypass from b 's supervisor, yielding the edge $a \rightarrow c$. The intermediate graph is now $a \rightarrow b$, $a \rightarrow c$, and $c \rightarrow d$.
2. We subsequently remove $c \rightarrow d$. Because c 's supervisor is now a , a bypass is generated from a to d .
3. The final target hierarchy, h' , consists of the edges $\{a \rightarrow b, a \rightarrow c, a \rightarrow d\}$.

If, instead, we reverse the sequence and remove $c \rightarrow d$ first:

1. The deletion of $c \rightarrow d$ triggers a bypass from c 's current supervisor, yielding the edge $b \rightarrow d$. The intermediate graph is now $a \rightarrow b \rightarrow c$ and $b \rightarrow d$.
2. We subsequently remove $b \rightarrow c$. Because b 's supervisor is a , a bypass is generated from a to c .
3. The final target hierarchy, h'' , consists of the edges $\{a \rightarrow b, a \rightarrow c, b \rightarrow d\}$.

Notice that this chain contains no overlapping reporting lines. Yet the exact same subset of removed edges produces two distinct organizational topologies. The order of operations matters because an early removal, such as $b \rightarrow c$, can “burn the bridge” needed to cascade a bypass further down the chain.

Path redundancy compounds this chronological interference by dynamically activating or suppressing bypass generation. Consider a matrix hierarchy where c has two supervisors, b and x , both of whom report to a , i.e., $a \rightarrow b \rightarrow c$ and $a \rightarrow x \rightarrow c$, and c supervises d , $c \rightarrow d$. Suppose we remove $b \rightarrow c$ and $c \rightarrow d$.

- If we remove $b \rightarrow c$ first, a does *not* receive a bypass to c , because it maintains an independent alternative path via x . When we subsequently remove $c \rightarrow d$, c 's remaining supervisor x receives a bypass $x \rightarrow d$.
- If we remove $c \rightarrow d$ first, both of c 's current supervisors, b and x , receive bypasses to d , yielding $b \rightarrow d$ and $x \rightarrow d$. When we subsequently remove $b \rightarrow c$, a again receives no bypass.

When matrix architecture allows multiple overlapping reporting paths, the chronological footprint of a bypass edge becomes ambiguous. If a target graph contains a bypass edge, it may be impossible to statically determine which alternative path was consumed to create it, or whether an alternative path suppressed its creation until a later step.

Because of this compounding chronological interference, evaluating whether a target hierarchy is obtainable from an initial DAG cannot in general be resolved by a simple static topological mapping. A static formula would have to evaluate the sequential contingencies of possible removal orders across intersecting paths. Rather than yielding structural insight, such a formula would obscure a brute-force algorithmic search beneath set-theoretic notation.

The reduction trace in [Definition 3](#) simply records this procedural reality. By explicitly tracking the sequence of intermediate topologies, it makes clear that the structural transformation of general DAGs is inherently chronological and cannot be algebraically compressed without losing its dynamic path dependencies.

3.2. Restoring geometric elegance: Multipath-free DAGs

The preceding examples show that chronological interference has two sources: the order-dependence of sequential removals, and the ambiguity introduced by overlapping alternative paths. It is tempting to assume that restricting the domain to organizations without redundant reporting lines restores commutativity. This is not correct. As demonstrated by the $a \rightarrow b \rightarrow c \rightarrow d$ chain example, chronological interference persists even in a simple, non-redundant chain of command. Removing $b \rightarrow c$ before $c \rightarrow d$ yields a different target hierarchy than removing $c \rightarrow d$ before $b \rightarrow c$. Thus removals do not commute.

What changes when redundant paths are banned is not the existence of chronological dependencies, but our ability to detect them.

Definition 6 (multipath-free hierarchy). A hierarchy $h \in \mathcal{H}_n$ is *multipath-free* if, for any two individuals $u, v \in V(h)$, there exists at most one directed path from u to v .

In a multipath-free DAG, lateral matrix reporting is permitted, but for any ordered pair of individuals there is at most one chain of command connecting the first to the second. Consequently, chronological dependencies do not result in a combinatorial family of alternative paths. Rather, they become statically visible along unique paths.

To build intuition for how these chronological dependencies become visible in the target graph, consider again the four-person chain h consisting of $a \rightarrow b \rightarrow c \rightarrow d$. Recall that removing the edges $\{b \rightarrow c, c \rightarrow d\}$ yielded two distinct target hierarchies depending on the sequence of operations:

- $h' = \{a \rightarrow b, a \rightarrow c, a \rightarrow d\}$, when $b \rightarrow c$ was removed first.
- $h'' = \{a \rightarrow b, a \rightarrow c, b \rightarrow d\}$, when $c \rightarrow d$ was removed first.

In a general DAG, if we were presented with h'' as a target, the potential existence of alternative matrix paths would make it impossible to determine statically whether the long bypass $a \rightarrow d$ is missing because of the chronological sequence of removals, or because bypasses were routed through some other path.

However, because h is multipath-free, the path

$$a \rightarrow b \rightarrow c \rightarrow d$$

is unique. Because it is unique, the static topology of the target graph reveals its chronological history. In h'' , the edge $b \rightarrow d$ is present. This presence records that $c \rightarrow d$ was removed while b was still connected to c , allowing b to survive as a direct supervisor of d . If a target graph were to validly contain the long bypass $a \rightarrow d$, the sequence of removals must have consumed the intermediate nodes, leaving neither b nor c as direct supervisors of d , as in h' .

Thus, the presence of $b \rightarrow d$ in h'' rules out the long bypass $a \rightarrow d$. Because paths are unique, the chronology is visible in the target graph itself.

Lemma 4 (closure of multipath-free DAGs). *If $h \in \mathcal{H}_n$ is multipath-free and h^+ is obtained from h by one subordination removal, then h^+ is multipath-free.*

Proof. Suppose h^+ is obtained from h by removing $j \rightarrow i$. Any bypass added by [Definition 2](#) has the form $p \rightarrow i$, where $p \rightarrow j \in E(h)$. Since $p \rightarrow j \rightarrow i$ is already a path from p to i in h , the multipath-free property implies that no path from p to i avoiding j exists. Thus any added bypass replaces the unique path segment $p \rightarrow j \rightarrow i$.

Acyclicity is preserved: if a bypass $p \rightarrow i$ created a directed cycle in h^+ , then h^+ would contain a path $i \rightsquigarrow p$. Replacing each bypass on that path by its original two-edge segment produces a path from i to p in h , which together with $p \rightarrow j \rightarrow i$ would form a cycle in h . Hence h^+ is acyclic.

Now suppose h^+ contains two distinct directed paths from some u to some v . Because every bypass edge added during the removal of $j \rightarrow i$ shares the same target node i , and because h^+ is acyclic, any directed path in h^+ can contain at most one bypass edge. Indeed, once a path enters i via one bypass edge, any second bypass edge would require the path to leave i and later return to i , creating a directed cycle through i .

If neither path contains a bypass, then both are distinct directed paths in h , contradicting multipath-freeness. Suppose instead that one or both paths contain a bypass $p \rightarrow i$. Expand each such bypass into the segment $p \rightarrow j \rightarrow i$. This expansion does not repeat any vertices. Indeed, the node j cannot appear earlier on the path, since the subpath $j \rightsquigarrow p$ together with $p \rightarrow j$ would form a cycle in h . Similarly, j cannot appear later on the path, since the subpath $i \rightsquigarrow j$ together with $j \rightarrow i$ would form a cycle in h . Therefore each expanded path is a simple directed path in h .

Expansion is injective on directed paths. Indeed, every bypass edge in h^+ has the unique form $p \rightarrow i$, and its expansion replaces that edge by the two-edge segment $p \rightarrow j \rightarrow i$, while no ordinary edge is altered. Hence the expanded path uniquely determines the original path, obtained by replacing each occurrence of $p \rightarrow j \rightarrow i$ with the corresponding bypass edge $p \rightarrow i$. Therefore two distinct paths in h^+ cannot expand to the same directed path in h .

Thus h would contain two distinct directed paths from u to v , contradicting multipath-freeness. Therefore h^+ is multipath-free. ■

This structural restriction allows the sequential subordination-removal process to be characterized by a one-shot geometric condition.

Definition 7 (static reduction). Let $h, h' \in \mathcal{H}_n$ be multipath-free hierarchies. We say that h' is a *static reduction* of h if there exists a bijection $\phi : V(h) \rightarrow V(h')$ such that:

(i) **Reachability Containment.** If $u' \rightsquigarrow v'$ in h' , then

$$\phi^{-1}(u') \rightsquigarrow \phi^{-1}(v')$$

in h .

(ii) **Unique-Path Consumption.** For every direct edge $(u', v') \in E(h')$, let

$$u = \phi^{-1}(u'), \quad v = \phi^{-1}(v').$$

If $(u, v) \notin E(h)$, then, by multipath-freeness, there is a unique directed path in h ,

$$u = d_0 \rightarrow d_1 \rightarrow \cdots \rightarrow d_m = v, \quad m \geq 2.$$

For every internal node $d_r, 1 \leq r \leq m-1$,

$$(\phi(d_r), v') \notin E(h').$$

Reachability Containment guarantees that no new line of authority is created. Unique-Path Consumption encodes the chronological bridge-burning constraint: if a bypass edge from u' to v' exists in the target graph, then none of the internal intermediaries on the original unique chain can survive as a direct supervisor of v' in the target graph.

Theorem 1. Let $h, h' \in \mathcal{H}_n$ be multipath-free hierarchies. Then $h \succ_H h'$ if and only if h' is a static topological reduction of h .

Proof. Necessity. Suppose $h \succ_H h'$. Then there exists a reduction trace from h to h' via some bijection $\phi : V(h) \rightarrow V(h')$. Let $\bar{h}'$ denote the pullback of h' to the node set $V(h)$, so that

$$E(\bar{h}') = E_\phi^* = \{(u, v) : (\phi(u), \phi(v)) \in E(h')\}.$$

By [Lemma 4](#), every intermediate graph appearing in the trace is multipath-free.

We prove by induction on the trace length L that $\bar{h}'$ is a static topological reduction of h via the identity map on $V(h)$; equivalently, h' is a static topological reduction of h via ϕ .

If $L = 0$, then $E(h) = E(\bar{h}')$, so h' is a relabeling of h , and both conditions of [Definition 7](#) hold immediately.

Assume the claim holds for traces of length $L - 1$. Let g be the penultimate intermediate hierarchy, and suppose the final transition removes the edge $j \rightarrow i \in E(g)$, producing $\bar{h}'$. By the induction hypothesis, g is a static topological reduction of h . The final step deletes $j \rightarrow i$ and adds only bypasses $p \rightarrow i$, where $p \rightarrow j \in E(g)$ and there is no path from p to i in g avoiding j .

Reachability Containment is preserved. Indeed, every newly added bypass $p \rightarrow i$ replaces the pre-existing path

$$p \rightarrow j \rightarrow i$$

in g . Therefore every path in $\bar{h}'$ can be expanded into a path in g . Since g is, by the induction hypothesis, reachability-contained in h , every reachability relation in $\bar{h}'$ is also present in h .

It remains to verify Unique-Path Consumption. Let $(u, v) \in E(\bar{h}')$ be an edge such that $(u, v) \notin E(h)$. Since h is multipath-free, there is a unique path

$$u = d_0 \rightarrow d_1 \rightarrow \cdots \rightarrow d_m = v, \quad m \geq 2,$$

from u to v in h . We must show that no internal node d_r , $1 \leq r \leq m - 1$, is an immediate supervisor of v in $\bar{h}'$.

There are two cases.

First, suppose that $(u, v) \in E(g)$, so the edge was already present before the final removal. By the induction hypothesis, no internal node on the unique u -to- v path in h is an immediate supervisor of v in g . The final transition only adds new incoming edges to the node i . Hence, if $v \neq i$, no new immediate supervisor of v is created, and the desired condition remains true.

Now suppose $v = i$. If some internal node d_r became an immediate supervisor of i in $\bar{h}'$ during the final transition, then $d_r \rightarrow i$ must be one of the newly added bypasses. Hence d_r is an immediate supervisor of j in g . Since d_r lies on the unique path from u to i in h , and $d_r \rightarrow j$ in g , reachability containment for g implies that $d_r \rightsquigarrow j \rightsquigarrow i$ is represented along the unique u -to- i path in h . Thus j is also an internal node on the unique path from u to i in h . But $j \rightarrow i \in E(g)$, so j was an immediate supervisor of i in g , contradicting the induction hypothesis applied to the edge $(u, i) \in E(g)$. Therefore no internal node of the unique u -to- i path in h becomes an immediate supervisor of i in $\bar{h}'$.

Second, suppose that $(u, v) \notin E(g)$. Then (u, v) must be newly created in the final transition. Hence $v = i$, $u = p$ for some immediate supervisor p of j in g , and the new edge $p \rightarrow i$ replaces the segment

$$p \rightarrow j \rightarrow i$$

in g . If $p \rightarrow j \notin E(h)$, the induction hypothesis applied to the direct edge $p \rightarrow j$ in g implies that no internal node on the unique path from p to j in h is an immediate supervisor of j in g ; if $p \rightarrow j \in E(h)$, there are no such internal nodes. By the same reasoning applied to the direct edge $j \rightarrow i$ in g , no internal node on the unique path from j to i in h is an immediate supervisor of i in g .

In the final transition, the node j is explicitly removed as an immediate supervisor of i . The only new incoming edges to i originate from immediate supervisors of j in g . Since no internal node on the unique path from p to j in h is an immediate supervisor of j in g , none of those internal nodes is promoted to an immediate supervisor of i . Similarly, no internal node on the unique path from j to i in h was an immediate supervisor of i in g ,

and the final step does not add incoming edges from such nodes. Finally, j itself is not an immediate supervisor of i in $\bar{h}'$, since $j \rightarrow i$ is the edge removed.

Thus no internal node on the combined unique path

$$p \rightsquigarrow j \rightsquigarrow i$$

in h is an immediate supervisor of i in $\bar{h}'$. This is exactly the Unique-Path Consumption condition for the newly created bypass edge $p \rightarrow i$.

Therefore $\bar{h}'$ is a static topological reduction of h , and hence h' is a static topological reduction of h via ϕ .

Sufficiency. Suppose h' is a static topological reduction of h via ϕ . We construct a reduction trace from h to h' via ϕ . Work on the node set $V(h)$, and write

$$E_\phi^* = \{(u, v) : (\phi(u), \phi(v)) \in E(h')\}.$$

Let g be the current intermediate hierarchy, initially $g = h$. We maintain the invariant that h' is a static topological reduction of g via the same bijection ϕ .

For any intermediate hierarchy g , define the surplus edge set

$$D(g) = E(g) \setminus E_\phi^*$$

and the associated multiset of depths

$$M_D(g) = \{s_g(x) : (x, y) \in D(g)\},$$

where $s_g(x)$ denotes the number of strict ancestors of x in g .

First suppose there exists a target edge $(u', v') \in E(h')$ whose preimage

$$(u, v) = (\phi^{-1}(u'), \phi^{-1}(v'))$$

is not an edge of g . Choose such an edge with v of maximal depth in g among all missing target preimages. Since g is multipath-free, Reachability Containment gives a unique path

$$u = d_0 \rightarrow d_1 \rightarrow \cdots \rightarrow d_m = v, \quad m \geq 2.$$

Each displayed arrow is an edge of g ; in particular,

$$d_{m-2} \rightarrow d_{m-1} \in E(g),$$

so d_{m-2} is an immediate supervisor of d_{m-1} in g . By Unique-Path Consumption,

$$(\phi(d_{m-1}), v') \notin E(h'),$$

so the terminal edge $d_{m-1} \rightarrow v$ belongs to $D(g)$. In particular, it is a surplus edge. Remove $d_{m-1} \rightarrow v$. By [Lemma 4](#), the resulting intermediate hierarchy remains multipath-free. By [Definition 2](#), a bypass $r \rightarrow v$ is added for each immediate supervisor r of d_{m-1} in g . In particular, the bypass

$$d_{m-2} \rightarrow v$$

is added, so the preimage path for (u', v') shortens by one edge.

We now verify that the static-reduction invariant is preserved. To establish Reachability Containment, it suffices to show that every target edge $(a', b') \in E(h')$ retains a representing path in the new intermediate graph. Let $(a, b) = (\phi^{-1}(a'), \phi^{-1}(b'))$. By the invariant, there is a representing path P from a to b in g . If P avoids the removed edge $d_{m-1} \rightarrow v$, it survives intact. Suppose P uses $d_{m-1} \rightarrow v$. Because g is multipath-free, P is the unique path from a to b in g .

Could $d_{m-1} \rightarrow v$ be the first edge of P ? If so, $a = d_{m-1}$. Since $d_{m-1} \rightarrow v \in D(g)$, it is not the preimage of any target edge, so $v \neq b$. Thus v is a strict ancestor of b in g , which means that b has strictly greater depth than v in g . If (a, b) were a missing target preimage, this would contradict our maximal-depth choice of v . Hence (a, b) must already be a direct edge in g . But since g is multipath-free, its unique path is exactly the length-one direct edge $d_{m-1} \rightarrow b$. Since $v \neq b$, this path avoids $d_{m-1} \rightarrow v$, a contradiction.

Therefore, $d_{m-1} \rightarrow v$ cannot be the first edge of P . It must be preceded by some immediate supervisor r on the path. After the removal, the segment

$$r \rightarrow d_{m-1} \rightarrow v$$

is replaced by the bypass

$$r \rightarrow v.$$

Thus the preimage path is shortened rather than destroyed. Since every target edge retains a representing path, any target reachability relation $x' \rightsquigarrow y'$ in h' remains represented by concatenating the representing paths of its constituent edges.

For Unique-Path Consumption, consider any target edge $(a', b') \in E(h')$ whose preimage is not direct after the removal. If its preimage path avoids $d_{m-1} \rightarrow v$, the condition is unchanged. If its preimage path uses $d_{m-1} \rightarrow v$, then the segment $r \rightarrow d_{m-1} \rightarrow v$ is replaced by $r \rightarrow v$. The removed node d_{m-1} no longer appears on the shortened path. Any remaining internal node was already internal on the old path and, by the invariant, did not map to an immediate supervisor of b' in h' . Thus no forbidden internal supervisor is introduced, and Unique-Path Consumption remains valid.

If every edge of h' already has a direct preimage in g , but $D(g) \neq \emptyset$, choose an edge $x \rightarrow y \in D(g)$ with y of maximal depth in g among targets of surplus edges, and remove it. Again, **Lemma 4** ensures that the resulting intermediate hierarchy remains multipath-free. By definition, the removed edge $x \rightarrow y$ is not the preimage of any target edge. Any bypass produced either coincides with a target preimage, in which case it preserves a required direct edge, or belongs to the surplus set and will be removed later. Since every target edge already has a direct preimage in g , no target path depends on a non-target bypass for its representation. Hence any newly created bypass that is not itself a target preimage is genuinely surplus. Because no direct preimage of a target edge is destroyed, every target edge continues to have a direct preimage in the new graph. Consequently, Reachability Containment is preserved, and Unique-Path Consumption is vacuous because there are no target edges lacking a direct preimage. Thus the static-reduction invariant is fully preserved.

In both cases above, the explicitly removed edge belongs to the surplus set $D(g)$, so its source depth is removed from $M_D(g)$. Every newly generated bypass that does not correspond to a target edge enters the surplus set. The source of any such bypass is an immediate supervisor of the removed edge's source, and therefore has a strictly smaller depth. Any generated bypass that does coincide with a target preimage simply drops out

of $D(g)$ altogether. Moreover, the operation creates no new reachability relations. Hence, for every node x , every strict ancestor of x in the new graph was already a strict ancestor of x in g , so the ancestor set of x can only shrink. Therefore all surviving source-depth values weakly decrease. By the definition of the strict multiset extension of $<$ on $\mathbb{N}$ —that is, replacing one or more entries by finitely many strictly smaller entries—replacing one occurrence of the removed source depth by any finite multiset of strictly smaller source depths, while weakly decreasing all remaining entries, yields a strict decrease in the multiset ordering. Hence each step strictly decreases $M_D(g)$, and termination follows.

At termination, no target edge lacks a direct preimage and no surplus edge remains. Hence the terminal edge set is exactly E_ϕ^* . The sequence of removals constructed above is therefore a reduction trace from h to h' via ϕ . Thus $h \succsim_H h'$. ■

3.3. Axiomatic characterization of fixed-size pre-orders

Having established the structural properties of hierarchical dominance via the reduction trace, we now characterize the class of pre-orders that are compatible with the core relation $\succsim_H$. We seek extensions of $\succsim_H$ that preserve its fundamental ordering properties while potentially introducing additional comparability between hierarchies.

We define two levels of consistency with respect to the core pre-order $\succsim_H$, mirroring our bifurcated Subordination Removal axioms.

Definition 8 ($\succsim_H$ -consistency). A hierarchical pre-order $\succsim$ on $\mathcal{H}_n$ is $\succsim_H$ -consistent if it satisfies the following conditions for all $h, h' \in \mathcal{H}_n$:

- (i) $h \sim_H h' \Rightarrow h \sim h'$.
- (ii) $h \succ_H h' \Rightarrow h \succ h'$.
- (iii) $h \succ h'$ whenever h' can be obtained from h by removing a non-redundant subordination relation.

Definition 9 (strong $\succsim_H$ -consistency). A hierarchical pre-order $\succsim$ on $\mathcal{H}_n$ is *strongly* $\succsim_H$ -consistent if it satisfies the following conditions for all $h, h' \in \mathcal{H}_n$:

- (i) $h \sim_H h' \Rightarrow h \sim h'$.
- (ii) $h \succ_H h' \Rightarrow h \succ h'$.

A strongly $\succsim_H$ -consistent pre-order preserves all strict comparisons and indifference classes of $\succsim_H$. Baseline $\succsim_H$ -consistency is more permissive: it allows a pre-order to register indifference when a removal is redundant and leaves the effective supervisory relation unchanged. Thus, baseline consistency treats hierarchy primarily through the effective chain of command rather than through formal organizational wiring.

Before establishing our extension theorems, we state an instrumental property of the core pre-order $\succsim_H$. The following lemma confirms that bidirectional weak dominance under $\succsim_H$ is structurally equivalent to an isomorphic relabeling.

Lemma 5. For any hierarchies $h, h' \in \mathcal{H}_n$, $h \sim_H h'$ implies that h is a relabeling of h' .

The proof of [Lemma 5](#) is deferred to [Appendix A.4](#). With this lemma and [Proposition 1](#), which established that $h >_H h'$ precisely characterizes sequential removals, we can now demonstrate the exact equivalence between our two levels of axiomatic strictness, [SR](#) and [SSR](#), and our two levels of consistency.

Theorem 2. *Let $\succsim$ be a hierarchical pre-order on $\mathcal{H}_n$.*

- (i) $\succsim$ satisfies [A](#) and [SR](#) if and only if it is $\succsim_H$ -consistent.
- (ii) $\succsim$ satisfies [A](#) and [SSR](#) if and only if it is strongly $\succsim_H$ -consistent.

Proof. *Proof of (i): Sufficiency.* Suppose $\succsim$ is $\succsim_H$ -consistent. If h' is a relabeling of h , then $h \sim_H h'$, implying $h \succsim_H h'$ and $h' \succsim_H h$. By consistency, $h \sim h'$, satisfying [A](#).

If h' is obtained from h by removing a subordination relation, [Lemma 2](#) establishes $h >_H h'$, which by consistency yields $h \succsim h'$. Furthermore, if the removed edge is non-redundant, condition (iii) of [Definition 8](#) guarantees $h > h'$. Thus, [SR](#) is satisfied.

Necessity. Suppose $\succsim$ satisfies [A](#) and [SR](#). We must establish $\succsim_H$ -consistency.

First, if $h \sim_H h'$, [Lemma 5](#) establishes that h is a relabeling of h' . Since $\succsim$ satisfies [A](#), $h \sim h'$.

Second, suppose $h >_H h'$. By [Proposition 1](#), h' can be obtained from some relabeling h^* of h by a finite sequence of subordination removals. Because $\succsim$ satisfies [SR](#), each removal yields a weak dominance, giving the chain

$$h^* \succsim h_L \succsim \cdots \succsim h_1 \succsim h'.$$

Since h^* is a relabeling of h and $\succsim$ satisfies [A](#), $h \sim h^*$. By transitivity, $h \succsim h'$.

Finally, [SR](#) explicitly dictates $h > h'$ for non-redundant removals. Thus, $\succsim$ is $\succsim_H$ -consistent.

Proof of (ii): Sufficiency. Suppose $\succsim$ is strongly $\succsim_H$ -consistent. If h' is a relabeling of h , then $h \sim_H h'$, and strong consistency implies $h \sim h'$, satisfying [A](#).

If h' is obtained from h by removing any subordination relation, [Lemma 2](#) establishes $h >_H h'$, and strong consistency enforces $h > h'$, satisfying [SSR](#).

Necessity. Suppose $\succsim$ satisfies [A](#) and [SSR](#). The proof for $h \sim_H h' \Rightarrow h \sim h'$ follows identically from [A](#) and [Lemma 5](#).

Suppose $h >_H h'$. By [Proposition 1](#), h' is obtained from a relabeling h^* of h by one or more removals. Because $\succsim$ satisfies [SSR](#), every single removal yields a strict dominance. Thus,

$$h^* > h_L > \cdots > h_1 > h'.$$

Since $h \sim h^*$ by Anonymity, transitivity implies $h > h'$. Thus, $\succsim$ is strongly $\succsim_H$ -consistent. ■

It is worth noting that the distinction between [SR](#) and [SSR](#)—and consequently between consistency and strong consistency—has no bite on the multipath-free domain. Indeed, multipath-freeness rules out redundant edges: if $j \rightarrow i$ is an edge, then the one-edge path $j \rightarrow i$ is already a directed path from j to i , so the existence of any distinct alternative path from j to i would violate multipath-freeness. Hence every subordination removal in a multipath-free hierarchy is non-redundant, and [SR](#) already forces a strict decrease in hierarchy. In particular, this applies to standard organizational trees, which form a special case of multipath-free hierarchies. It is only on the full domain of directed acyclic

graphs, where matrix-style multiple reporting lines may generate redundant edges, that the bifurcation between **SR** and **SSR** becomes substantive.

We now turn to cardinal measures. A hierarchical index on $\mathcal{H}_n$ is a function $I : \mathcal{H}_n \rightarrow \mathbb{R}$ that assigns a hierarchical degree $I(h)$ to every hierarchy. Each index I induces a complete hierarchical order $\succsim_I$ on $\mathcal{H}_n$ defined by

$$h \succsim_I h' \iff I(h) \geq I(h').$$

For any hierarchy $h \in \mathcal{H}_n$, let $s_h(i)$ denote the number of strict ancestors, direct and indirect supervisors, of individual i in h . We define the supervisor index I_S as

$$I_S(h) = \frac{1}{n} \sum_{i \in V(h)} s_h(i).$$

The complete order $\succsim_{I_S}$ evaluates hierarchy entirely through reachability. If h' is obtained from h by a non-redundant subordination removal, then the transitive chain of command is strictly reduced, and therefore $I_S(h) > I_S(h')$. If instead the removed edge is redundant, then the transitive supervisory relation is unchanged, so $I_S(h) = I_S(h')$. Thus $\succsim_{I_S}$ satisfies **SR** but need not satisfy **SSR** when redundant edges are present in the domain (for example, whenever $n \geq 3$). Since I_S depends only on the numbers of strict ancestors, it is invariant under relabeling and therefore satisfies **A**.

Proposition 3. *The hierarchical order $\succsim_{I_S}$ defined on $\mathcal{H}_n$ is complete, reflexive, and transitive, and satisfies **A** and **SR**.*

By **Proposition 3** and **Theorem 2(i)**, $\succsim_{I_S}$ is $\succsim_H$ -consistent. On any fixed-size domain $\mathcal{H}_n$ containing redundant edges, however, $\succsim_{I_S}$ fails **SSR**, because removing a redundant edge leaves I_S unchanged.

To confirm that the class of strongly $\succsim_H$ -consistent pre-orders is non-empty, consider the following index. For $d \in \{0, \dots, n-1\}$, define

$$\omega_d(h) = |\{(x, y) \in E(h) : s_h(x) = d\}|,$$

the number of formal reporting lines whose source has exactly d strict ancestors. Define the formal wiring profile

$$\Omega_n(h) = (\omega_{n-1}(h), \omega_{n-2}(h), \dots, \omega_0(h)).$$

Profiles are compared lexicographically from higher source depths to lower source depths.

Define the lexicographic hierarchical order $\succsim_{SSR}$ on $\mathcal{H}_n$ by

$$h \succsim_{SSR} h' \iff (I_S(h), \Omega_n(h)) \geq_{\text{lex}} (I_S(h'), \Omega_n(h')).$$

If h' is obtained from h by a non-redundant subordination removal, then the transitive chain of command is strictly reduced, so $I_S(h) > I_S(h')$. If the removed edge is redundant, then the transitive supervisory relation is unchanged, so $I_S(h) = I_S(h')$. However, in this case the formal wiring profile strictly decreases. Indeed, removing $j \rightarrow i$ deletes one edge whose source has depth $s_h(j)$, while every bypass edge $p \rightarrow i$ has source p satisfying $p \rightarrow j$, and therefore

$$s_h(p) < s_h(j).$$

Moreover, the operation creates no new reachability relation, so no surviving edge source can move to a higher depth class: all surviving source depths weakly decrease. Let d^* be the highest depth at which the profile changes. Then all coordinates above d^* are unchanged, and at depth d^* the profile must strictly decrease, because one edge with source depth $s_h(j)$ is removed while every newly created bypass contributes only at strictly lower source depths. Hence

$$\Omega_n(h) >_{\text{lex}} \Omega_n(h')$$

when the profiles are compared from higher source depths to lower source depths. Therefore $h >_{SSR} h'$ for every subordination removal.

Thus $\succ_{SSR}$ is complete, reflexive, and transitive, satisfies **A** and **SSR**, and therefore, by **Theorem 2(ii)**, constitutes a complete strongly $\succ_H$ -consistent extension on $\mathcal{H}_n$.

4. Comparing hierarchies of varying sizes

The extension of our analysis to hierarchies of varying sizes can be accomplished through the Replication Principle, following the methodology established in [Carbonell-Nicolau \(2025\)](#). This principle allows us to compare hierarchical structures across different organizational scales while preserving the essential properties of the hierarchical orders. The formal procedure involves creating comparable replicated structures and then applying our fixed-size results to these standardized hierarchies.

The set of all hierarchies of any positive size is defined as

$$\mathcal{H} = \bigcup_{n=1}^{\infty} \mathcal{H}_n,$$

where $\mathcal{H}_n$ is the set of all n -person hierarchies.

Definition 10. A *hierarchical pre-order* on $\mathcal{H}$ is a reflexive and transitive binary relation on $\mathcal{H}$.

For a positive integer k , let h^k denote the hierarchy consisting of k independent copies of h :

$$h^k = \underbrace{(h, \dots, h)}_{k \text{ times}}.$$

We call h^k a *replication* of h . By convention, $h^1 = h$.

The Replication Principle asserts that replicating a hierarchy does not change its hierarchical degree.

Replication Principle (RP). A hierarchical pre-order $\succ$ on $\mathcal{H}$ satisfies **RP** if for any two hierarchies h and h' in $\mathcal{H}$, $h' \sim h$ whenever h' is a replication of h .

We extend the fixed-size core relation $\succ_H$ to $\mathcal{H}$ by comparing common-size replications. If $h \in \mathcal{H}_m$ and $h' \in \mathcal{H}_n$, define

$$h \succ_H h' \iff h^n \succ_H (h')^m$$

under the fixed-size relation on $\mathcal{H}_{mn}$.

The following observation records that the fixed-size relation $\succsim_H$ is compatible with replication.

Lemma 6. *Let $g, g' \in \mathcal{H}_N$. If $g \succsim_H g'$ on $\mathcal{H}_N$, then for every positive integer r , $g^r \succsim_H (g')^r$ on $\mathcal{H}_{rN}$.*

Proof. Suppose $g \succsim_H g'$. Then there exists a bijection $\phi : V(g) \rightarrow V(g')$ and a reduction trace from g to g' via ϕ . Replicate this trace independently on each of the r copies of g , using the replicated bijection ϕ^r that acts as ϕ on each copy. Since the copies are pairwise disconnected, each removal step acts within a single copy and does not affect the others. Hence the replicated sequence is a reduction trace from g^r to $(g')^r$ via ϕ^r . Therefore $g^r \succsim_H (g')^r$. ■

Lemma 7. *The hierarchical pre-order $\succsim_H$ defined on $\mathcal{H}$ is reflexive and transitive and satisfies **A**, **SR**, **SSR**, and **RP**.*

Proof. Reflexivity. If $h \in \mathcal{H}_m$, then $h^m \succsim_H h^m$ by reflexivity of the fixed-size relation on $\mathcal{H}_{m^2}$. Hence $h \succsim_H h$ on $\mathcal{H}$.

Transitivity. Let $h \in \mathcal{H}_m$, $h' \in \mathcal{H}_n$, and $h'' \in \mathcal{H}_p$, and suppose

$$h \succsim_H h' \quad \text{and} \quad h' \succsim_H h''.$$

By definition,

$$h^n \succsim_H (h')^m \quad \text{on } \mathcal{H}_{mn},$$

and

$$(h')^p \succsim_H (h'')^n \quad \text{on } \mathcal{H}_{np}.$$

By **Lemma 6**, replicating the first relation p times and the second relation m times gives

$$h^{np} \succsim_H (h')^{mp} \quad \text{and} \quad (h')^{pm} \succsim_H (h'')^{nm}$$

on $\mathcal{H}_{mnp}$. Since $(h')^{mp} = (h')^{pm}$, transitivity of the fixed-size relation yields

$$h^{np} \succsim_H (h'')^{nm}$$

on $\mathcal{H}_{mnp}$. Therefore $h \succsim_H h''$ on $\mathcal{H}$.

Anonymity. If h' is a relabeling of h and both belong to $\mathcal{H}_m$, then $(h')^m$ is a relabeling of h^m . By Anonymity of the fixed-size core relation,

$$h^m \sim_H (h')^m$$

on $\mathcal{H}_{m^2}$. Hence $h \sim_H h'$ on $\mathcal{H}$, so **A** holds.

SR and SSR. Suppose h' is obtained from $h \in \mathcal{H}_m$ by removing a single subordination relation. Then $(h')^m$ is obtained from h^m by performing the same removal independently in each copy. By **Proposition 1**, this yields $h^m \succ_H (h')^m$ on $\mathcal{H}_{m^2}$. Hence $h \succ_H h'$ on $\mathcal{H}$. In particular, the weak inequality required by **SR** holds for every removal, and the strict inequality required by **SSR** also holds for every removal.

RP. If $h' = h^k$ is a replication of $h \in \mathcal{H}_m$, then $h' \in \mathcal{H}_{km}$, and by definition

$$h \succsim_H h' \quad \Leftrightarrow \quad h^{km} \succsim_H (h^k)^m.$$

But both h^{km} and $(h^k)^m$ consist of km independent copies of h , so they are relabelings of one another. Hence

$$h^{km} \sim_H (h^k)^m,$$

which gives $h \succsim_H h'$. The same argument in reverse gives $h' \succsim_H h$. Therefore $h \sim_H h'$, establishing **RP**. ■

For hierarchical pre-orders defined on the extended domain $\mathcal{H}$, we use the same two consistency notions as in **Definitions 8** and **9**. That is, $\succsim_H$ -consistency and strong $\succsim_H$ -consistency are defined by the same implications, with $\mathcal{H}_n$ replaced by $\mathcal{H}$, so that all quantifiers now range over $h, h' \in \mathcal{H}$.

The following result extends **Theorem 2** to the domain $\mathcal{H}$.

Theorem 3. *Let $\succsim$ be a hierarchical pre-order on $\mathcal{H}$.*

(i) *$\succsim$ satisfies **A**, **SR**, and **RP** if and only if it is $\succsim_H$ -consistent.*

(ii) *$\succsim$ satisfies **A**, **SSR**, and **RP** if and only if it is strongly $\succsim_H$ -consistent.*

Proof. Proof of (i): Sufficiency. Suppose $\succsim$ is $\succsim_H$ -consistent. If h' is a relabeling of h , then $h \sim_H h'$ by **Lemma 7**. By consistency, $h \sim h'$, so **A** holds. If h' is obtained from h by removing a subordination relation, then **Lemma 7** gives $h \succ_H h'$, and consistency therefore yields $h \succ h'$. If the removed edge is non-redundant, condition (iii) of **Definition 8** gives $h \succ h'$. Hence **SR** holds. Finally, if $h' = h^k$ is a replication of h , then $h \sim_H h'$ by **Lemma 7**, so consistency gives $h \sim h'$. Thus **RP** holds.

Necessity. Suppose $\succsim$ satisfies **A**, **SR**, and **RP**. We must prove $\succsim_H$ -consistency.

Fix $h \in \mathcal{H}_m$ and $h' \in \mathcal{H}_n$, and define

$$h_r = h^n \quad \text{and} \quad h'_r = (h')^m.$$

Then $h_r, h'_r \in \mathcal{H}_{mn}$. By **RP** for $\succsim$,

$$h \sim h_r \quad \text{and} \quad h' \sim h'_r.$$

If $h \sim_H h'$, then by definition of the extended core relation, $h_r \sim_H h'_r$ on $\mathcal{H}_{mn}$. Since $h_r, h'_r \in \mathcal{H}_{mn}$, **Lemma 5** implies that h_r is a relabeling of h'_r . By **A** for $\succsim$, $h_r \sim h'_r$. Together with $h \sim h_r$ and $h' \sim h'_r$, transitivity yields $h \sim h'$.

If $h \succ_H h'$, then by definition of the extended core relation, $h_r \succ_H h'_r$ on $\mathcal{H}_{mn}$. Since $h_r, h'_r \in \mathcal{H}_{mn}$, **Proposition 1** implies that h'_r can be obtained from some relabeling h_r^* of h_r by one or more subordination removals. Because $\succsim$ satisfies **SR**, each removal yields a weak dominance, giving

$$h_r^* \succsim h_L \succsim \cdots \succsim h_1 \succsim h'_r.$$

Since h_r^* is a relabeling of h_r and $\succsim$ satisfies **A**, we have $h_r \sim h_r^*$, hence $h_r \succsim h'_r$. Using $h \sim h_r$ and $h'_r \sim h'$, transitivity yields $h \succsim h'$.

Finally, if h' is obtained from h by removing a non-redundant subordination relation, then **SR** explicitly gives $h \succ h'$. Therefore $\succsim$ is $\succsim_H$ -consistent.

Proof of (ii): Sufficiency. Suppose $\succsim$ is strongly $\succsim_H$ -consistent. If h' is a relabeling of h , then $h \sim_H h'$ by **Lemma 7**, and strong consistency implies $h \sim h'$. Thus **A** holds. If h' is obtained from h by removing any subordination relation, then **Lemma 7** gives $h \succ_H h'$,

and strong consistency yields $h > h'$. Thus **SSR** holds. If $h' = h^k$ is a replication of h , then $h \sim_H h'$ by **Lemma 7**, and strong consistency gives $h \sim h'$. Hence **RP** holds.

Necessity. Suppose $\succsim$ satisfies **A**, **SSR**, and **RP**. We must prove strong $\succsim_H$ -consistency.

The argument for

$$h \sim_H h' \implies h \sim h'$$

is identical to part (i).

Now suppose $h >_H h'$. By definition of the extended core relation, $h_r >_H h'_r$ on $\mathcal{H}_{mn}$. Since $h_r, h'_r \in \mathcal{H}_{mn}$, **Proposition 1** implies that h'_r can be obtained from a relabeling h_r^* of h_r by one or more subordination removals. Because $\succsim$ satisfies **SSR**, every single removal yields a strict dominance. Thus

$$h_r^* > h_L > \dots > h_1 > h'_r.$$

Since h_r^* is a relabeling of h_r and $\succsim$ satisfies **A**, we have $h_r \sim h_r^*$, and hence $h_r > h'_r$. Using $h \sim h_r$ and $h'_r \sim h'$, we obtain

$$h \succ h_r > h'_r \succ h',$$

so $h \succ h'$. If $h' \succ h$, then $h'_r \succ h_r$ by $h' \sim h'_r$, $h \sim h_r$, and transitivity, contradicting $h_r > h'_r$. Therefore $h > h'$. This proves strong $\succsim_H$ -consistency. ■

The hierarchical pre-order $\succsim_S$ introduced in **Carbonell-Nicolau (2025)** can be extended to $\mathcal{H}$ analogously. For $h \in \mathcal{H}_m$ and $h' \in \mathcal{H}_n$, define

$$h \succsim_S h' \iff h^n \succsim_S (h')^m$$

under the fixed-size relation on $\mathcal{H}_{mn}$. By the same replication argument used above, this extension is reflexive and transitive and satisfies **A**, **SR**, and **RP**. Therefore, by **Theorem 3(i)**, $\succsim_S$ is $\succsim_H$ -consistent. However, exactly as in the fixed-size domain, $\succsim_S$ evaluates hierarchy entirely through reachability and fails **SSR**. Therefore it is not strongly $\succsim_H$ -consistent. As demonstrated in **Carbonell-Nicolau (2025)**, certain hierarchical structures remain incomparable under $\succsim_H$, whereas $\succsim_S$ induces a consistent ordering.

The hierarchical index I_S defined previously can be naturally extended to $\mathcal{H}$. For any hierarchy $h \in \mathcal{H}$ with $n = |V(h)|$ individuals, define

$$I_S(h) = \frac{1}{n} \sum_{i \in V(h)} s_h(i),$$

where $s_h(i)$ denotes the number of supervisors of individual i in h . The hierarchical order induced by I_S on $\mathcal{H}$ is defined as

$$h \succsim_{I_S} h' \iff I_S(h) \geq I_S(h').$$

Proposition 4. *The hierarchical order $\succsim_{I_S}$ defined on $\mathcal{H}$ is complete, reflexive, and transitive, and satisfies axioms **A**, **SR**, and **RP**.*

Proof. Completeness, reflexivity, and transitivity are immediate because $\succsim_{I_S}$ is induced by the real-valued index I_S .

To verify **A**, observe that relabeling preserves the number of strict ancestors of every node, and hence preserves the average value $I_S(h)$. Therefore relabelings are indifferent under $\succsim_{I_S}$.

To verify **RP**, let h^k be a replication of h . Replicating h multiplies both the total number of supervisors and the number of individuals by k , so $I_S(h^k) = I_S(h)$. Hence $h^k \sim_{I_S} h$.

To verify **SR**, let h' be obtained from h by removing a subordination relation. This operation creates no new ancestor-descendant reachability relation. Hence

$$s_{h'}(i) \leq s_h(i) \quad \text{for all } i \in V(h).$$

If the removed edge is non-redundant, at least one reachability relation is lost, so the inequality is strict for some node, and therefore $I_S(h) > I_S(h')$. If the removed edge is redundant, the transitive supervisory relation is unchanged, so $I_S(h) = I_S(h')$. Thus $\succsim_{I_S}$ satisfies **SR**. ■

By **Proposition 4** and **Theorem 3(i)**, $\succsim_{I_S}$ is $\succsim_H$ -consistent. Since $\mathcal{H}$ contains hierarchies with redundant edges, $\succsim_{I_S}$ fails **SSR** and is not strongly $\succsim_H$ -consistent.

To construct a strongly $\succsim_H$ -consistent complete extension on the variable-size domain, we normalize the formal wiring profile by population size. For $h \in \mathcal{H}$, let $n_h = |V(h)|$, and for each $d \geq 0$ define

$$\bar{\omega}_d(h) = \frac{1}{n_h} |\{(x, y) \in E(h) : s_h(x) = d\}|.$$

Only finitely many coordinates are nonzero. Let

$$\bar{\Omega}(h) = (\dots, \bar{\omega}_2(h), \bar{\omega}_1(h), \bar{\omega}_0(h))$$

denote the resulting normalized formal wiring profile, compared lexicographically from the highest nonzero source-depth coordinate downward.

Define the lexicographic hierarchical order $\succsim_{SSR}$ on $\mathcal{H}$ by

$$h \succsim_{SSR} h' \quad \Leftrightarrow \quad (I_S(h), \bar{\Omega}(h)) \geq_{\text{lex}} (I_S(h'), \bar{\Omega}(h')).$$

Replication preserves I_S . It also preserves $\bar{\Omega}$, since replicating a hierarchy k times multiplies both the number of edges of each source-depth type and the population size by k . Hence $\succsim_{SSR}$ satisfies **RP**. Relabeling preserves both I_S and $\bar{\Omega}$, so $\succsim_{SSR}$ satisfies **A**.

If h' is obtained from h by a non-redundant subordination removal, then $I_S(h) > I_S(h')$. If the removed edge is redundant, then $I_S(h) = I_S(h')$, but the normalized formal wiring profile strictly decreases. Indeed, the removed edge has source depth $s_h(j)$, while any bypass generated by **Definition 2** has source an immediate supervisor of j , and hence has strictly smaller depth. Moreover, the operation creates no new reachability relation, so no surviving edge source can move to a higher depth class. Let d^* be the highest depth at which the profile changes. Then all coordinates above d^* are unchanged, and at depth d^* the coordinate strictly decreases, because one edge is removed at some depth while every newly created bypass contributes only to strictly lower depths. Therefore

$$\bar{\Omega}(h) >_{\text{lex}} \bar{\Omega}(h').$$

Thus $h \succ_{SSR} h'$ for every subordination removal. Consequently, $\succ_{SSR}$ satisfies **A**, **SSR**, and **RP**. Since it is defined lexicographically from complete real-valued / ordered profile data, it is also complete, reflexive, and transitive. Therefore, by **Theorem 3(ii)**, $\succ_{SSR}$ is a complete strongly $\succ_H$ -consistent extension on $\mathcal{H}$.

5. Concluding remarks

Viewing hierarchy as an organizational channel through which economic power and resources are allocated shifts attention from abstract economic forces to concrete institutional structures that can be formally represented, measured, and compared.

We propose an axiomatic approach to measuring hierarchy in directed acyclic graphs, rooted in fundamental principles: Anonymity, Subordination Removal, Strong Subordination Removal, and the Replication Principle. The transition from traditional organizational trees to modern matrix structures introduces substantial topological complexity. We demonstrate that in general directed acyclic graphs, the structural flattening of an organization is inherently path-dependent due to chronological interference. Because of the failure of commutativity and the ambiguity introduced by path redundancy, determining whether one hierarchy can be reduced to another requires tracing the explicit chronological sequence of subordination removals.

Our analysis identifies the precise boundary at which chronological complexity gives way to geometric tractability. On the multipath-free domain, chronological dependencies do not disappear into commutativity; instead, they become statically visible along unique paths. As a result, the existence of a valid sequence of removals admits a one-shot geometric characterization based on Reachability Containment and Unique-Path Consumption.

Furthermore, the transition to matrix structures necessitates a bifurcation of the subordination-removal concept itself. By distinguishing between baseline Subordination Removal and Strong Subordination Removal, our dual-consistency framework clarifies how different measures treat formal reporting redundancy. Baseline Subordination Removal evaluates hierarchy through the effective, transitive chain of command, while Strong Subordination Removal treats even redundant formal reporting lines as strict contributors to hierarchical weight.

The core hierarchical pre-order developed in this paper, $\succ_H$, captures the unambiguous comparisons implied by sequential subordination removal. This pre-order establishes a minimal consistency threshold, enabling the development of more complete rankings, such as those derived from the average number of supervisors. The framework then extends naturally from fixed-size hierarchies to hierarchies of varying sizes through the Replication Principle.

The link between this general framework and earnings inequality remains a promising area for future exploration. By integrating our generalized measurement approach with the well-established field of inequality measurement, future work can investigate how complex hierarchical structures influence income distribution—particularly in contexts where upward mobility in earnings aligns with advancement through organizational matrices. This perspective opens a path for connecting axiomatic hierarchy measurement with broader questions about organizational design, compensation structures, and systemic economic inequality.

A. Appendix

A.1. Proof of Lemma 1

Lemma 1. *The binary relation $\succ_H$ defined on $\mathcal{R}_n$ is reflexive and transitive, and satisfies A.*

Proof. Reflexivity. Set $L = 0$, $E^0 = E(h)$, and let ϕ be the identity bijection. Then the conditions of Definition 3 are trivially satisfied, so $h \succ_H h$.

Anonymity. If h' is a relabeling of h , let $\phi : V(h) \rightarrow V(h')$ be the relabeling bijection. Taking $L = 0$ gives

$$E^0 = E(h) = E_\phi^*,$$

so $h \succ_H h'$. Similarly, $h' \succ_H h$.

Transitivity. Suppose $h \succ_H h'$ via $\phi_1 : V(h) \rightarrow V(h')$, and $h' \succ_H h''$ via $\phi_2 : V(h') \rightarrow V(h'')$.

For any edge set $A \subseteq V(h') \times V(h')$, we use the standard pullback notation:

$$\phi_1^{-1}(A) = \{(\phi_1^{-1}(a), \phi_1^{-1}(b)) : (a, b) \in A\}.$$

Let

$$(e_1, \dots, e_{L_1}; E^0, \dots, E^{L_1})$$

be a reduction trace from h to h' via ϕ_1 , so

$$E^0 = E(h), \quad E^{L_1} = \phi_1^{-1}(E(h')).$$

Let

$$(f_1, \dots, f_{L_2}; F^0, \dots, F^{L_2})$$

be a reduction trace from h' to h'' via ϕ_2 , so

$$F^0 = E(h'), \quad F^{L_2} = \phi_2^{-1}(E(h'')).$$

For each $\ell = 0, \dots, L_2$, define the pulled-back edge set

$$\tilde{F}^\ell = \phi_1^{-1}(F^\ell),$$

and for each $\ell = 1, \dots, L_2$, if $f_\ell = (a_\ell, b_\ell)$, define the pulled-back removed edge

$$\tilde{f}_\ell = (\phi_1^{-1}(a_\ell), \phi_1^{-1}(b_\ell)).$$

Because relabeling preserves immediate-supervisor relations and preserves the existence of directed paths avoiding a specified node, Definition 2 commutes with relabeling. Hence for each ℓ , $\tilde{f}_\ell \in \tilde{F}^{\ell-1}$, and $\tilde{F}^\ell$ is obtained from $\tilde{F}^{\ell-1}$ by removing $\tilde{f}_\ell$ and adding exactly the bypass edges prescribed by Definition 2. Thus

$$(\tilde{f}_1, \dots, \tilde{f}_{L_2}; \tilde{F}^0, \dots, \tilde{F}^{L_2})$$

is a reduction trace on $V(h)$. Moreover,

$$\tilde{F}^0 = \phi_1^{-1}(F^0) = \phi_1^{-1}(E(h')) = E^{L_1},$$

so it begins exactly where the first trace ends. Its terminal edge set is

$$\widetilde{F}^{L_2} = \phi_1^{-1}(F^{L_2}) = \phi_1^{-1}(\phi_2^{-1}(E(h''))) = (\phi_2 \circ \phi_1)^{-1}(E(h'')).$$

Therefore, concatenating the sequence of edges

$$e_1, \dots, e_{L_1}, \widetilde{f}_1, \dots, \widetilde{f}_{L_2}$$

and the sequence of edge sets

$$E^0, \dots, E^{L_1} = \widetilde{F}^0, \widetilde{F}^1, \dots, \widetilde{F}^{L_2}$$

yields a reduction trace from h to h'' via $\phi_2 \circ \phi_1$. Hence $h \succ_H h''$. ■

A.2. Proof of Lemma 3

Lemma 3. *If there is a nonempty reduction trace from h to h' , then*

$$M_{\text{src}}(h') < M_{\text{src}}(h).$$

Proof. It suffices to show that a single subordination removal strictly decreases the source-depth multiset. Let g^+ be obtained from an intermediate hierarchy g by removing the edge $j \rightarrow i$. Removing $j \rightarrow i$ deletes one instance of $s_g(j)$ from $M_{\text{src}}(g)$.

Every bypass edge added by **Definition 2** has the form $p \rightarrow i$, where $p \rightarrow j \in E(g)$. Thus p is a strict ancestor of j in g , and hence

$$s_g(p) < s_g(j).$$

Moreover, each bypass edge $p \rightarrow i$ replaces the pre-existing directed path

$$p \rightarrow j \rightarrow i.$$

Hence the operation creates no new ancestor-descendant reachability relation: indeed, any directed path in g^+ that uses a bypass edge $p \rightarrow i$ can be expanded in g by replacing $p \rightarrow i$ with the path $p \rightarrow j \rightarrow i$, while every other edge of g^+ is already an edge of g . Therefore every reachability relation in g^+ was already present in g . Consequently,

$$s_{g^+}(v) \leq s_g(v) \quad \text{for all } v \in V(g).$$

Therefore, the removed source-depth value $s_g(j)$ is replaced only by values

$$s_{g^+}(p) \leq s_g(p) < s_g(j),$$

while all surviving source-depth values weakly decrease. By the definition of the strict multiset extension of $<$ on $\mathbb{N}$, replacing one occurrence of $s_g(j)$ by any finite multiset of strictly smaller values, while weakly decreasing all remaining entries, yields a strict decrease in the multiset ordering. Hence $M_{\text{src}}(g^+) < M_{\text{src}}(g)$.

Iterating this strict descent over the nonempty reduction trace and using transitivity of the multiset order gives $M_{\text{src}}(h') < M_{\text{src}}(h)$. ■

A.3. Proof of Lemma 2

Lemma 2. For $h, h' \in \mathcal{H}_n$, if h' is obtained from h by a single subordination removal in the sense of Definition 2, then $h \succ_H h'$.

Proof. Suppose h' is obtained from h by removing the subordination relation $j \rightarrow i$. Then, via the identity bijection on $V(h)$, this single removal yields a reduction trace of length $L = 1$, with removed-edge list $(j \rightarrow i)$ and edge sets

$$E^0 = E(h) \quad \text{and} \quad E^1 = E(h').$$

Hence $h \succ_H h'$.

By Lemma 3, this nonempty reduction trace implies

$$M_{\text{src}}(h') < M_{\text{src}}(h).$$

We claim that $h' \not\succ_H h$. Suppose, toward a contradiction, that $h' \succ_H h$.

If the reduction trace from h' to h is nonempty, then Lemma 3 gives

$$M_{\text{src}}(h) < M_{\text{src}}(h'),$$

contradicting the asymmetry of the strict multiset order.

If the reduction trace from h' to h is empty, then h' is a relabeling of h . Relabeling preserves ancestor relations and hence source depths, so

$$M_{\text{src}}(h') = M_{\text{src}}(h),$$

again contradicting

$$M_{\text{src}}(h') < M_{\text{src}}(h).$$

Therefore $h' \not\succ_H h$. Since $h \succ_H h'$ and $h' \not\succ_H h$, we conclude that $h \succ_H h'$. ■

A.4. Proof of Lemma 5

Lemma 5. For any hierarchies $h, h' \in \mathcal{H}_n$, $h \sim_H h'$ implies that h is a relabeling of h' .

Proof. Suppose $h \sim_H h'$. By definition,

$$h \succ_H h' \quad \text{and} \quad h' \succ_H h.$$

Thus, there exists a reduction trace from h to h' , and a reduction trace from h' to h .

Assume, for contradiction, that the reduction trace from h to h' is nonempty. By Lemma 3,

$$M_{\text{src}}(h') < M_{\text{src}}(h).$$

Now consider the reduction trace from h' to h . If this reduction trace is nonempty, then Lemma 3 gives

$$M_{\text{src}}(h) < M_{\text{src}}(h'),$$

contradicting the asymmetry of the strict multiset ordering. If this reduction trace is empty, then h is a relabeling of h' , so

$$M_{\text{src}}(h) = M_{\text{src}}(h'),$$

again contradicting $M_{\text{src}}(h') < M_{\text{src}}(h)$.

Therefore, the reduction trace from h to h' must be empty. By **Definition 3**, an empty reduction trace, $L = 0$, implies that

$$E(h) = E^0 = E_{\phi}^* = \{(u, v) : (\phi(u), \phi(v)) \in E(h')\}.$$

Equivalently,

$$(u, v) \in E(h) \Leftrightarrow (\phi(u), \phi(v)) \in E(h').$$

Thus h' is a relabeling of h via the bijection ϕ . ■

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